

General Formula for Continuous Probability Distribution

1. Probability Density Function (PDF):

- The PDF $f(x)$ defines the likelihood of the random variable X taking on a specific value within its range.
- For a continuous random variable:

$$P(X = x) = 0$$

because the probability of a continuous variable taking an exact value is zero; instead, probabilities are calculated over intervals.

2. Cumulative Distribution Function (CDF):

- The CDF $F(x)$ gives the probability that X is less than or equal to a certain value

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

3. Probability for an Interval:

- The probability of X falling within an interval $[a, b]$:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

When to Apply Specific Continuous Distributions

1. Uniform Distribution:

- **When?** The random variable is equally likely to occur anywhere within a range.
- **PDF:**

$$f(x) = \frac{1}{b - a}, \quad a \leq x \leq b$$

- **Applications:** Flight times within a fixed range, equally distributed temperatures over a day.

2. Normal Distribution:

- **When?** The data is symmetric and bell-shaped, where most values cluster around the mean.
- **PDF:**

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- μ : Mean (center of the distribution)
- σ : Standard deviation (spread of the distribution)
- **Applications:** Heights, test scores, measurement errors.

3. Exponential Distribution:

- **When?** The random variable measures the time until the next event in a Poisson process (events occur independently and at a constant rate).
- **PDF:**

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- λ : Rate parameter (inverse of the mean).
- **Applications:** Time between arrivals of customers, lifetime of electronic components.

Example 6.4: Given a random variable X having a normal distribution with $\mu = 50$ and $\sigma = 10$, find the probability that X assumes a value between 45 and 62.

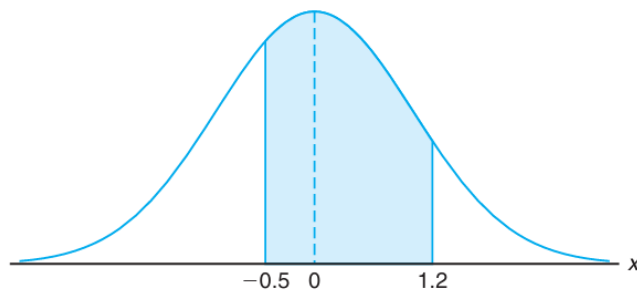


Figure 6.11: Area for Example 6.4.

Solution: The z values corresponding to $x_1 = 45$ and $x_2 = 62$ are

$$z_1 = \frac{45 - 50}{10} = -0.5 \text{ and } z_2 = \frac{62 - 50}{10} = 1.2.$$

Therefore,

$$P(45 < X < 62) = P(-0.5 < Z < 1.2).$$

$P(-0.5 < Z < 1.2)$ is shown by the area of the shaded region in Figure 6.11. This area may be found by subtracting the area to the left of the ordinate $z = -0.5$ from the entire area to the left of $z = 1.2$. Using Table A.3, we have

$$\begin{aligned} P(45 < X < 62) &= P(-0.5 < Z < 1.2) = P(Z < 1.2) - P(Z < -0.5) \\ &= 0.8849 - 0.3085 = 0.5764. \end{aligned}$$

Example 6.8: An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours.

Solution: The distribution of light bulb life is illustrated in Figure 6.15. The z values corresponding to $x_1 = 778$ and $x_2 = 834$ are

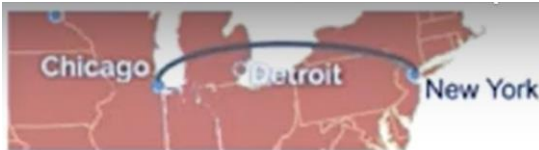
$$z_1 = \frac{778 - 800}{40} = -0.55 \text{ and } z_2 = \frac{834 - 800}{40} = 0.85.$$

Hence,

$$\begin{aligned} P(778 < X < 834) &= P(-0.55 < Z < 0.85) = P(Z < 0.85) - P(Z < -0.55) \\ &= 0.8023 - 0.2912 = 0.5111. \end{aligned}$$

where we let X be the flight time in minutes of an airplane traveling from Chicago to New York and the sample space of X is between 120 and 140 minutes this means the flight may arrive anytime between 120 and 140 minutes the flight time is equally likely to be any time within that interval and the flight time cannot be any time outside that range.

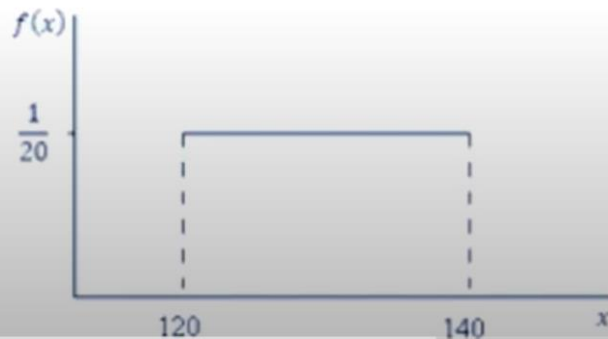
- calculate the probability that the flight time will be between 120 and 130 minute.
- calculate the probability that the flight time will be between 125 and 150 minute.
- calculate the 75th percentile.
- Calculate expected mean & variance.



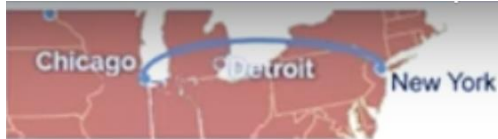
Let x = the flight time (in minutes)
an airplane travelling from Chicago
to New York

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{elsewhere} \end{cases}$$

$$= \begin{cases} \frac{1}{140-120} = \frac{1}{20} & \text{for } 120 \leq x \leq 140 \\ 0 & \text{elsewhere} \end{cases}$$



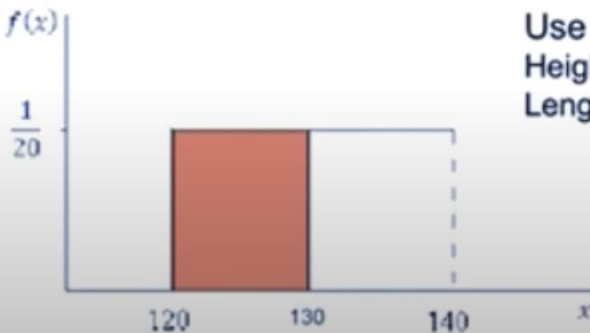
Activ
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Let x = the flight time (in minutes) of an airplane travelling from Chicago to New York

$$f(x) = \begin{cases} \frac{1}{140-120} = \frac{1}{20} & \text{for } 120 \leq x \leq 140 \\ 0 & \text{elsewhere} \end{cases}$$

Calculate the probability that the flight time will be between 120 and 130 minutes.

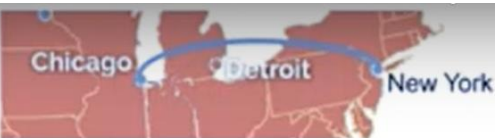


Use area of a rectangle (length $\times$ height)

Height = $f(x) = 0.05$

Length = $\Delta x = 130 - 120 = 10$

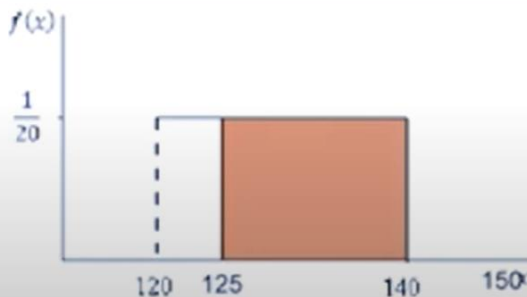
$$\begin{aligned} P(120 < x < 130) &= P(120 \leq x \leq 130) \\ &= \Delta x \cdot f(x) \\ &= (130 - 120) \left(\frac{1}{20} \right) \\ &= (10) \left(\frac{1}{20} \right) = 0.5 \end{aligned}$$



Let x = the flight time (in minutes) of an airplane travelling from Chicago to New York

$$f(x) = \begin{cases} \frac{1}{140-120} = \frac{1}{20} & \text{for } 120 \leq x \leq 140 \\ 0 & \text{elsewhere} \end{cases}$$

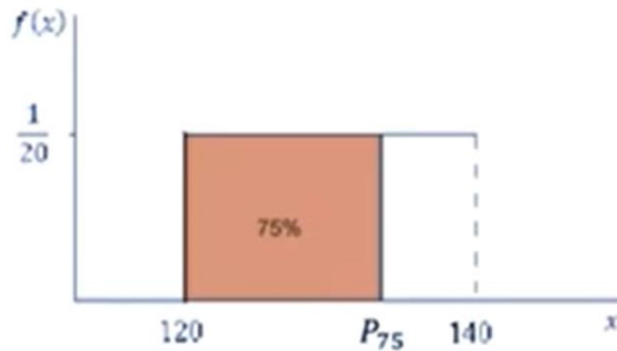
Calculate the probability that the flight time will be between 125 and 150 minutes.



$$\begin{aligned} P(125 < x < 150) &= P(125 < x < 140) + P(140 < x < 150) \\ &= \Delta x \cdot f(x) + \Delta x \cdot f(x) \\ &= (15) \left(\frac{1}{20} \right) + (10)(0) \\ &= 0.75 \end{aligned}$$

Note: Part of the range has an probability of 0

Calculate the 75th percentile



$$P(120 < x < P_{75}) = 0.75$$

$$(P_{75} - 120) \left(\frac{1}{20} \right) = 0.75$$

$$P_{75} - 120 = (0.75)(20)$$

$$P_{75} = 135$$

$$E(x) = \mu = \frac{a+b}{2}$$

$$\text{var}(x) = \sigma^2 = \frac{(b-a)^2}{12}$$



$$f(x) = \begin{cases} \frac{1}{140-120} = \frac{1}{20} & \text{for } 120 \leq x \leq 140 \\ 0 & \text{elsewhere} \end{cases}$$

$$E(x) = \frac{a+b}{2} = \frac{120+140}{2} = 130$$

$$\text{Var}(x) = \frac{(b-a)^2}{12} = \frac{(140-120)^2}{12} = 33.\bar{3}$$

- What is hypothesis testing, and what have you learned about its application?
- Explain how hypothesis testing is used in computer science with relevant examples.

What is Hypothesis Testing, and What Have You Learned About Its Application?

Hypothesis testing is a statistical method used to determine whether there is enough evidence in a sample to support or reject a claim about a population parameter. It involves comparing observed data to a theoretical assumption (null hypothesis) to check for statistical significance.

- Key Steps in Hypothesis Testing:

1. Formulate the null hypothesis (H_0) and the alternative hypothesis (H_1).
2. Choose the appropriate test statistic (e.g., z-test, t-test, chi-square).
3. Compute the test statistic from the sample data.
4. Compare the test statistic with the critical value at a given significance level (α).
5. Make a decision: Reject or fail to reject.

What I learned:

Hypothesis testing provides a structured framework to make data-driven decisions and validate claims with statistical evidence.

2. How is Hypothesis Testing Used in Computer Science?

Hypothesis testing is widely used in computer science to validate models, compare algorithms, and improve system performance. Some common applications include:

- 1. Algorithm Performance Comparison:**
 - **Example:** Test whether Algorithm A has a faster runtime than Algorithm B.
 - **Approach:** Use a t-test to compare the mean runtimes of the two algorithms.
- 2. A/B Testing in Software Development:**
 - **Example:** Test whether a new webpage layout (A) results in higher click-through rates than the old layout (B).
 - **Approach:** Use a proportion z-test to analyze user behavior.
- 3. Machine Learning Model Validation:**
 - **Example:** Test whether a new machine learning model performs better than a baseline model.
 - **Approach:** Use paired t-tests to compare the models' accuracies on test data.
- 4. Network Traffic Analysis:**
 - **Example:** Test whether the observed traffic pattern matches the expected behavior to detect anomalies.
 - **Approach:** Use a chi-square goodness-of-fit test.
- 5. Human-Computer Interaction (HCI):**
 - **Example:** Test whether users prefer a new interface design over the old one.
 - **Approach:** Collect survey data and apply a chi-square test for user preferences.
- 6. Error Rate Analysis in Systems:**
 - **Example:** Test if a new server configuration reduces the error rate below 1%.
 - **Approach:** Use a one-sample proportion test.

Conclusion:

Hypothesis testing allows computer scientists to make informed decisions, optimize processes, and ensure robust system performance based on statistical evidence.

QUESTION a random sample 100 recorded that in the United State during the past year show average life spent 71.8 assuming a population standard deviation of 8.9 does the same to indicate the mean life span spend today is greater than 70 year use? 0.05 use a Label of significance (the table value 0.9783).

Step 1: State the Hypotheses

We are testing if the average life expectancy today is greater than 70 years.

- Null Hypothesis (H_0): The mean life expectancy is 70 years.

$$H_0 : \mu = 70$$

- Alternative Hypothesis (H_a): The mean life expectancy is greater than 70 years.

$$H_a : \mu > 70$$

This is a **one-tailed test**.

Step 2: Identify the Given Data

- Sample mean ($\bar{x}$) = 71.8
- Population standard deviation (σ) = 8.9
- Hypothesized mean (μ_0) = 70
- Sample size (n) = Not given explicitly. Assuming n is available, we proceed symbolically.
- Significance level (α) = 0.05

Step 3: Choose the Test Statistic

Since the population standard deviation (σ) is known, we use the **z-test** formula:

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

Step 4: Calculate the Test Statistic

1. Plug in the values:

$$z = \frac{71.8 - 70}{\frac{8.9}{\sqrt{100}}}$$

2. Simplify the denominator:

$$\frac{\sigma}{\sqrt{n}} = \frac{8.9}{\sqrt{100}} = \frac{8.9}{10} = 0.89$$

3. Calculate z :

$$z = \frac{71.8 - 70}{0 \downarrow} = \frac{1.8}{0.89} \approx 2.02$$

Step 6: Make a Decision

- Compare z -value to z_{crit} :
 - Calculated $z = 2.02$
 - Critical $z_{\text{crit}} = 1.645$
- Since $z = 2.02$ is greater than $z_{\text{crit}} = 1.645$, we reject the null hypothesis.

Chi-Square Goodness-of-Fit Test: Explanation

The **chi-square goodness-of-fit test** is a statistical method used to determine whether a set of observed data matches a theoretical or expected distribution. It is commonly applied in scenarios where you want to assess whether a categorical variable follows a specified distribution.

Purpose of the Test

The test helps to answer the question:

- **Does the observed data fit the expected distribution?**

For example:

- Is a die fair (each face equally likely)?
- Are preferences for different brands as expected?
- Does a coin toss produce equal probabilities of heads and tails?

Key Concepts

1. Observed Frequencies (O_i):

- These are the actual counts observed in each category during the experiment.

2. Expected Frequencies (E_i):

- These are the theoretical counts expected in each category if the null hypothesis is true.
- Calculated as:

$$E_i = n \cdot p_i$$

where n is the total number of observations and p_i is the expected proportion for category i .

3. Chi-Square Statistic (χ^2):

- The test statistic measures the difference between observed and expected frequencies:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

- A larger value of χ^2 indicates greater deviation from the expected distribution.

4. Degrees of Freedom (df):

- The degrees of freedom are calculated as:

$$df = k - 1$$

where k is the number of categories.

5. Significance Level (α):

- The threshold for deciding whether to reject the null hypothesis. Common choices are $\alpha = 0.05$ or $\alpha = 0.01$.

6. Critical Value:

- Obtained from the chi-square distribution table based on the degrees of freedom and significance level.

Hypotheses

- **Null Hypothesis (H_0):** The coin is fair, meaning $P(\text{head}) = P(\text{tail}) = 0.5$.
- **Alternative Hypothesis (H_a):** The coin is not fair, meaning $P(\text{head}) \neq 0.5$.

Let's calculate the chi-square statistic step by step clearly. The formula is:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where:

- O_i : Observed frequency
- E_i : Expected frequency

We have the following data:

- Observed heads (O_H) = 63
- Observed tails (O_T) = 37
- Expected heads (E_H) = 50



Step 1: Calculate the differences ($O_i - E_i$) for heads and tails

- Heads: $O_H - E_H = 63 - 50 = 13$
- Tails: $O_T - E_T = 37 - 50 = -13$

Step 2: Square the differences $(O_i - E_i)^2$

- Heads: $(13)^2 = 169$
- Tails: $(-13)^2 = 169$

Step 3: Divide the squared differences by their expected values

- Heads: $\frac{169}{50} = 3.38$
- Tails: $\frac{169}{50} = 3.38$

Step 4: Sum the results

$$\chi^2 = 3.38 + 3.38 = 6.76$$

Final Chi-Square Value

$$\chi^2 = 6.76$$

Comparison with Critical Value

- Degrees of freedom $(df) = k - 1 = 2 - 1 = 1$
- Significance level $(\alpha = 0.05)$: Critical value from the chi-square table for $df = 1$ is **3.841**.

Decision

- If $\chi^2 > 3.841$, reject H_0 .
- Here, $\chi^2 = 6.76 > 3.841$.

10.81 A die is tossed 180 times with the following results:

x	1	2	3	4	5	6
f	28	36	36	30	27	23

Is this a balanced die? Use a 0.01 level of significance.

Step 1: State the hypotheses

- Null Hypothesis (H_0): The die is balanced, so all outcomes are equally likely.
- Alternative Hypothesis (H_a): The die is not balanced, so at least one outcome has a different probability.

Step 2: Set the level of significance

The significance level is $\alpha = 0.01$.

Step 3: Calculate expected frequencies

If the die is fair, the expected frequency for each face is:

$$\text{Expected frequency} = \frac{\text{Total tosses}}{\text{Number of outcomes}} = \frac{180}{6} = 30$$

where O_i is the observed frequency, and E_i is the expected frequency.

Outcome (x)	Observed (O_i)	Expected (E_i)	$(O_i - E_i)^2 / E_i$
1	28	30	$(28 - 30)^2 / 30 = 0.1333$
2	36	30	$(36 - 30)^2 / 30 = 1.2$
3	36	30	$(36 - 30)^2 / 30 = 1.2$
4	30	30	$(30 - 30)^2 / 30 = 0$
5	27	30	$(27 - 30)^2 / 30 = 0.3$
6	23	30	$(23 - 30)^2 / 30 = 1.6333$

$$\chi^2 = 0.1333 + 1.2 + 1.2 + 0 + 0.3 + 1.6333 = 4.4666$$

Step 5: Determine the critical value

The degrees of freedom for this test are:

$$df = \text{Number of outcomes} - 1 = 6 - 1 = 5$$

Using a chi-square distribution table at $\alpha = 0.01$ and 5 degrees of freedom, the critical value is:

$$\chi^2_{\text{critical}} = 15.086$$

Step 6: Compare and conclude

- If $\chi^2 \leq \chi^2_{\text{critical}}$, fail to reject H_0 .
- If $\chi^2 > \chi^2_{\text{critical}}$, reject H_0 .

Here, $\chi^2 = 4.4666$ and $\chi^2_{\text{critical}} = 15.086$. Since $4.4666 < 15.086$, we fail to reject H_0 .

Conclusion

At the 0.01 level of significance, there is insufficient evidence to conclude that the die is not balanced. It appears to be fair.